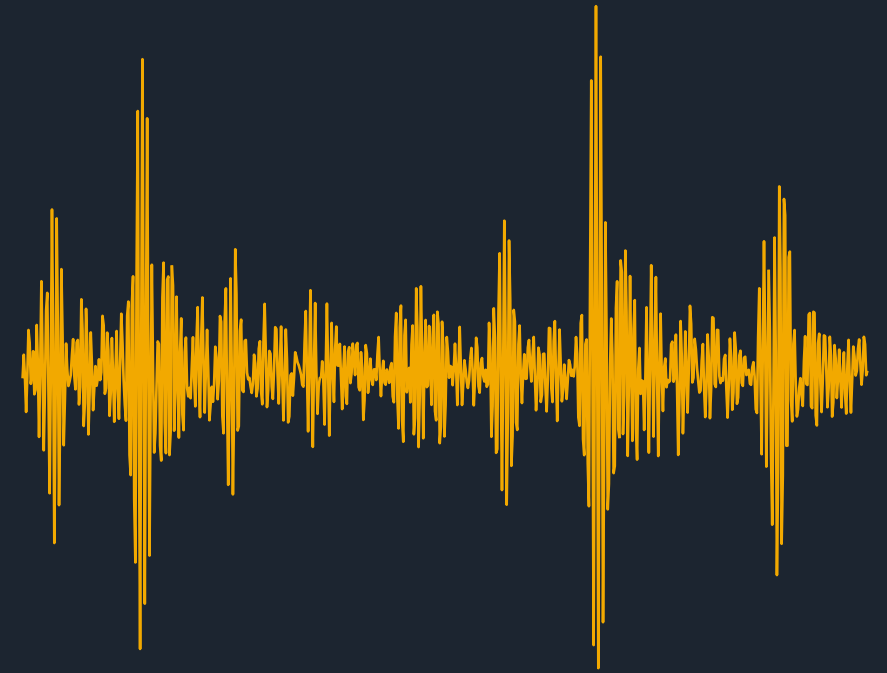


AI-Based Motor Vibration Fault Detection for Industries

Detecting bearing fault type and severity from vibration signals using
Machine Learning



Real drive-end vibration: inner race fault (0.021")

Why bearing health matters



Motors run every industry

Pumps, fans, compressors and conveyors all depend on electric motors running without interruption.



Bearings carry the load

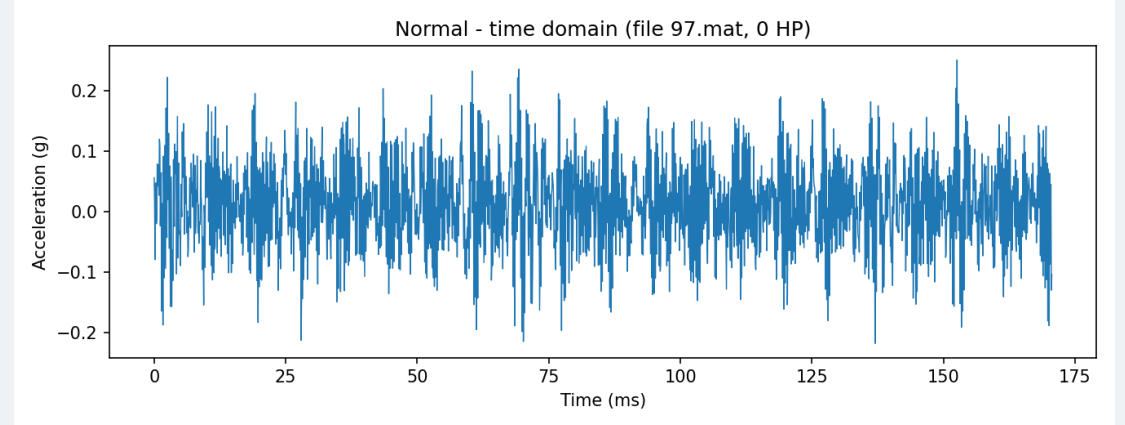
Every motor shaft spins on ball bearings. Cracks in the races or balls develop with wear and grow over time.



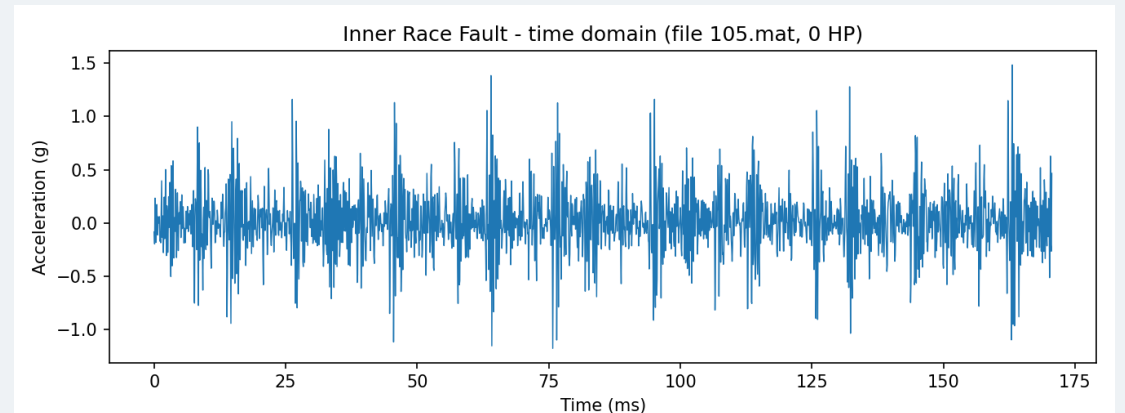
Damage changes the vibration

Each time a ball passes a crack it creates an impact. The vibration signal carries a signature of the fault.

Healthy bearing



Inner race fault



Problem statement



Needs an expert

Reading vibration spectra by hand requires a trained analyst and takes time for every machine.



Faults found too late

Without regular analysis, a small crack can grow until the motor fails unexpectedly.



"Faulty" isn't enough

Maintenance teams need to know which part is damaged and how badly, to plan the repair.

GOAL

Take a vibration signal and automatically report the bearing condition (**Normal, Inner Race, Ball or Outer Race fault**), the fault size, and a confidence score, with a clear warning when the result is uncertain.

Dataset: CWRU Bearing Data Center

40

recordings

2331

signal windows

12 kHz

sampling rate

0–3 HP

motor loads

Case Western Reserve University test rig: a 2 HP motor with an accelerometer on the drive-end bearing. Cracks of three sizes were made in the bearing on purpose and the vibration was recorded at four motor loads.



Data check we made

The Normal files were recorded at 48 kHz, not 12 kHz (shaft-speed peaks appeared at ¼ of the true frequency). We downsampled them to 12 kHz so the model learns the bearing condition, not a sampling-rate artefact.

Condition	0.007"	0.014"	0.021"	Windows
Normal	97–100			206
Inner Race Fault	105–108	169–172	209–212	709
Ball Fault	118–121	185–188	222–225	708
Outer Race Fault	130–133	197–200	234–237	708

CWRU file numbers; each group of 4 files = loads 0, 1, 2, 3 HP. Windows = 2048-sample segments (≈0.17 s).

Proposed solution

01



Signal processing

Convert to 12 kHz and cut each recording into 2048-sample windows.

02



Feature extraction

14 numbers per window: 9 time-domain and 5 frequency-domain (FFT) features.

03



Two ML models

Random Forest #1 finds the fault type. Random Forest #2 estimates the crack size.

04



Web application

Upload a CSV and see the waveform, spectrum, prediction, confidence and status.



Built to be robust: trained with extra noisy copies of every signal, accepts any sampling rate, and flags unfamiliar signals as "uncertain" instead of guessing.

System architecture



The same feature function is used for training and in the app, so the model always sees identical features.

Features and algorithm

Time domain (9)

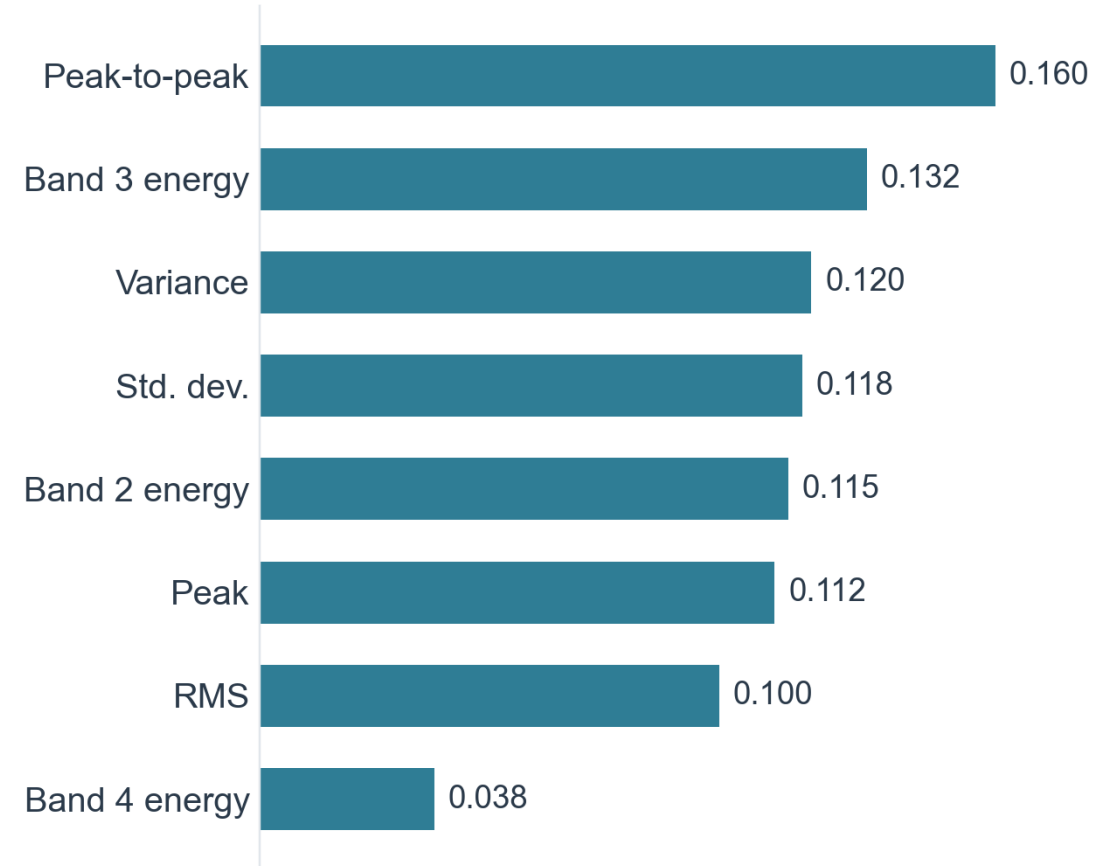
Mean	RMS	Variance
Std. dev.	Peak	Peak-to-peak
Crest factor	Kurtosis	Skewness

Frequency domain (5)

Dominant freq.	Band 1 energy	Band 2 energy
Band 3 energy	Band 4 energy	

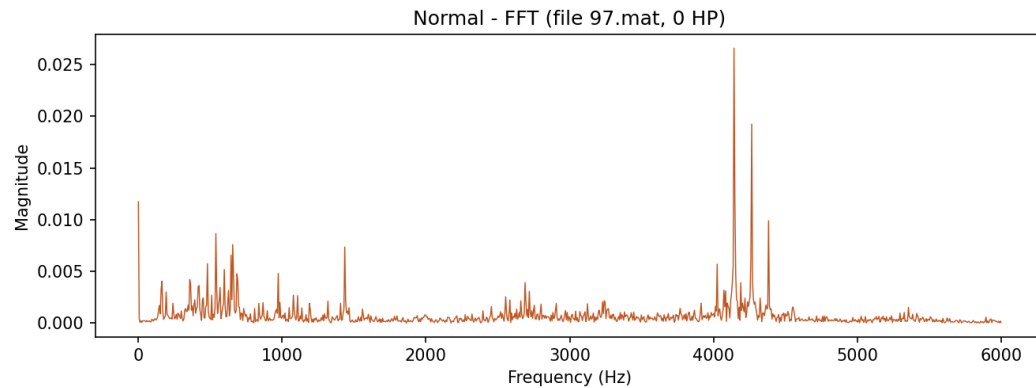
Model: Random Forest, 100 trees, balanced class weights, random_state = 42. Trained on clean windows + one noisy copy each (SNR 0–20 dB).

Feature importance (top 8, final model)

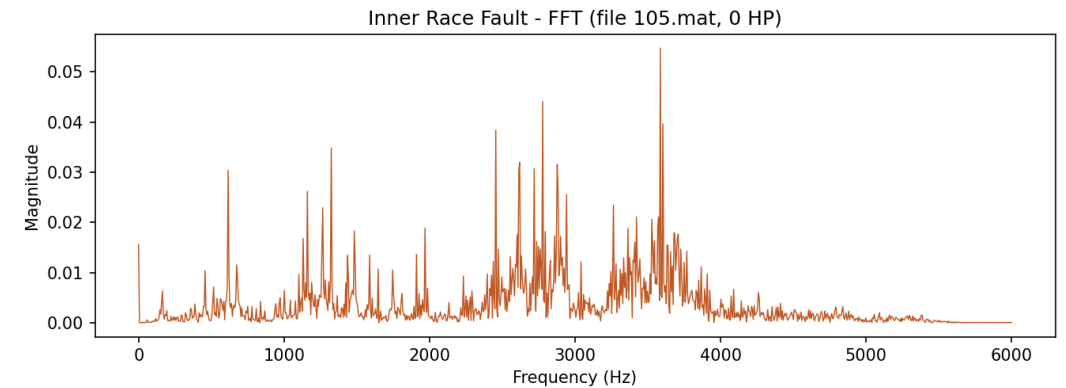


Each fault has its own frequency signature

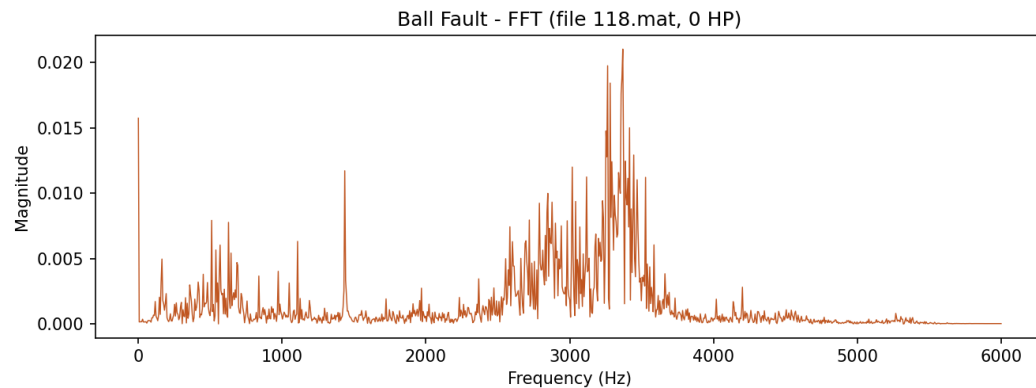
Normal



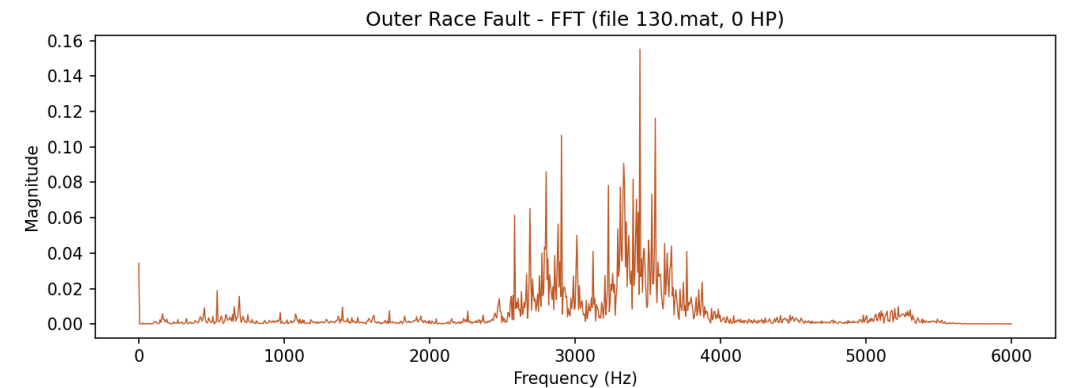
Inner race fault



Ball fault



Outer race fault



FFT spectra of the first 2048 samples of each 0 HP recording (0.007" faults). Faults move the vibration energy into different frequency bands.

Results: accuracy on unseen data

FAULT TYPE

99.57%

Random 80/20 split · 467 test windows

FAULT TYPE

98.82%

Train 0–2 HP, test unseen 3 HP · 591 windows

FAULT SIZE

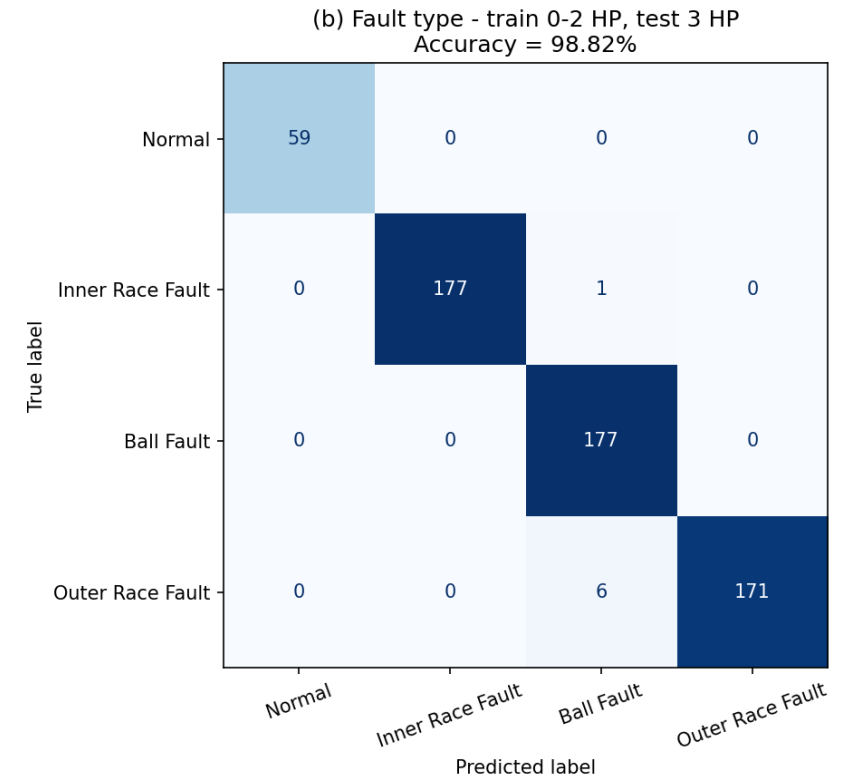
99.76%

Random 80/20 split · 425 test windows

FAULT SIZE

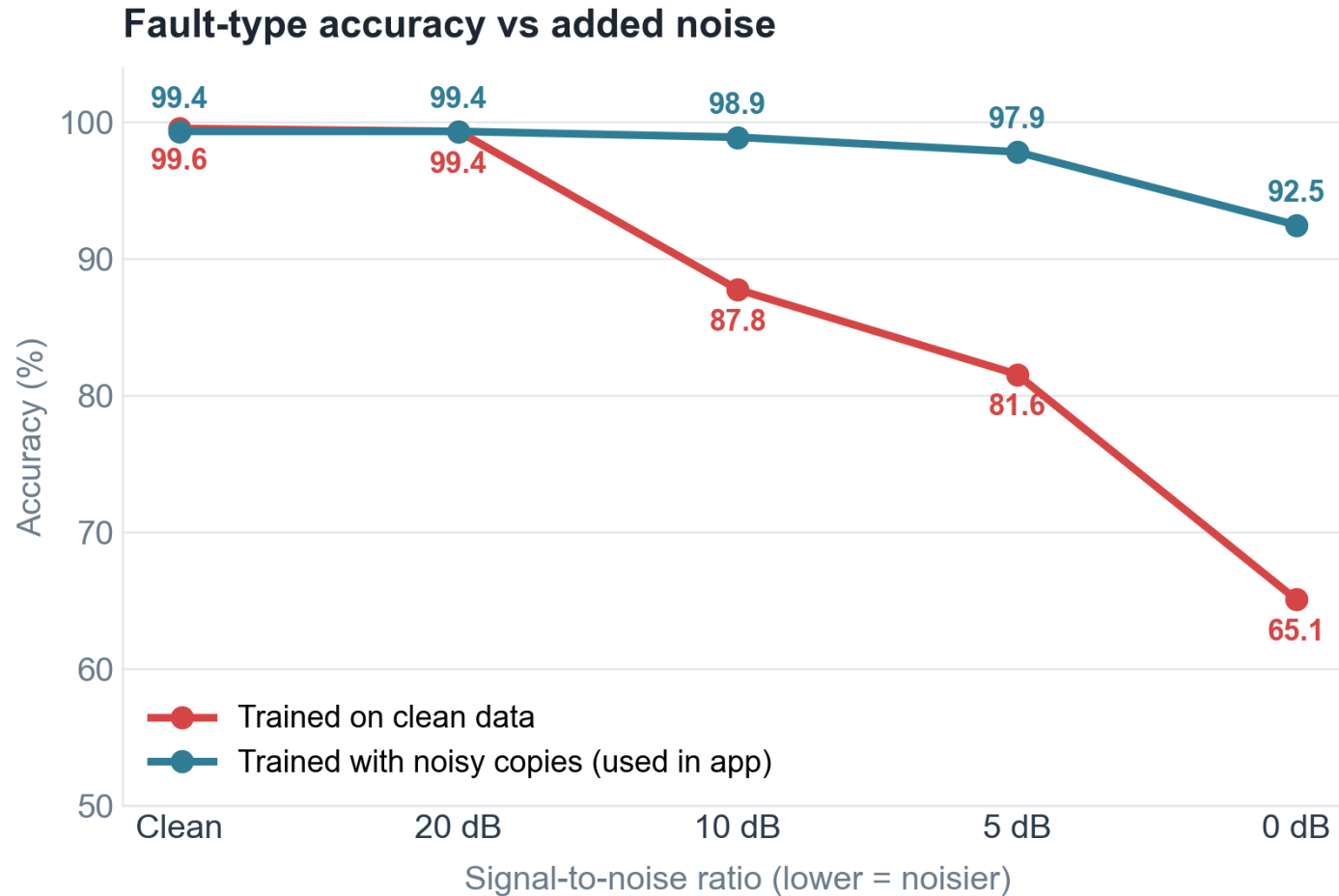
99.25%

Train 0–2 HP, test unseen 3 HP · 532 windows



Normal detected perfectly (59/59) at the unseen load.

Robust to noisy signals



AT 0 dB (NOISE = SIGNAL)

Before: 65.1%

92.5%

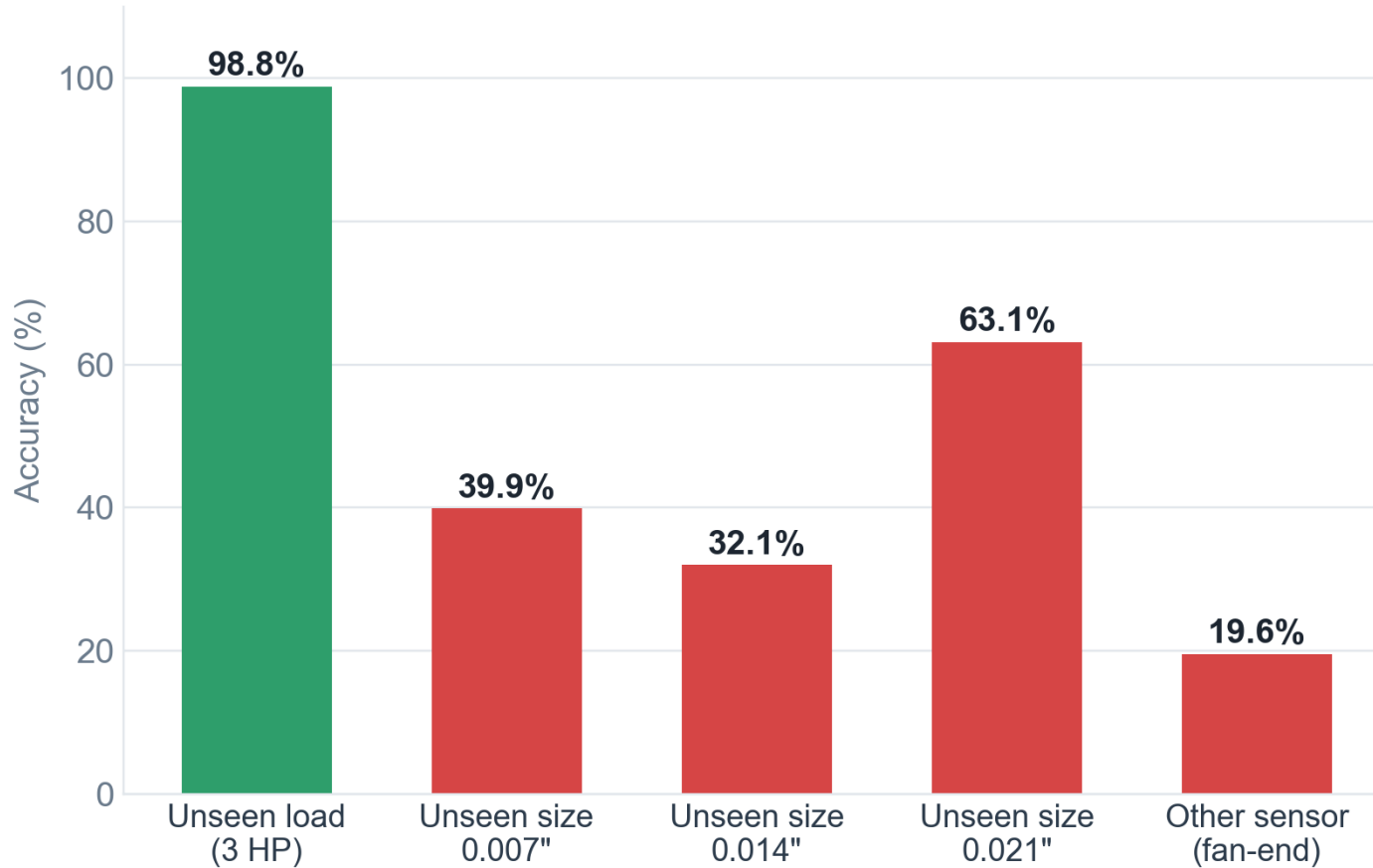
accuracy after training with noisy copies

How we tested: random noise added to the 467 test windows at decreasing signal-to-noise ratios.

Trade-off: only 0.21 points lower on clean data, so the noise-trained model is used in the app.

Where the model struggles

Fault-type accuracy when test conditions differ from training



Generalises across load

Trained on 0–2 HP, still 98.8% at 3 HP.



Not to new crack sizes

Leaving one size out of training drops accuracy to 32.1–63.1%.



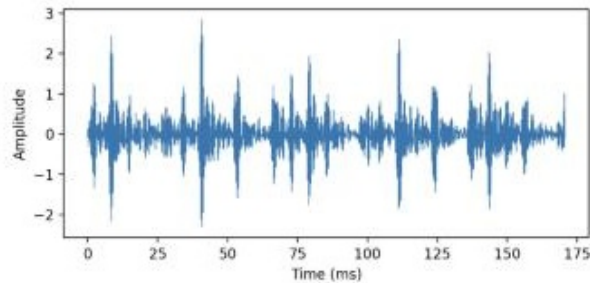
Not to a new sensor position

Drive-end model on fan-end data: 19.6%.

Lesson: train on data from the same fault sizes and sensor position the system will be used on.

Live demo: Streamlit web app

Vibration waveform



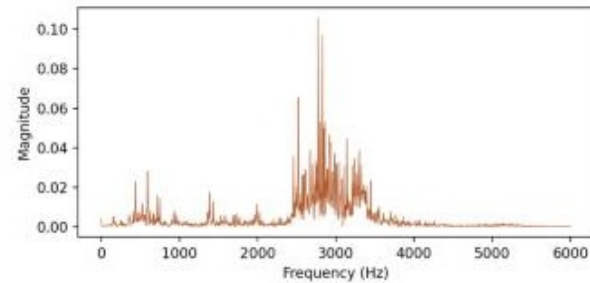
RMS

0.4376

Peak

2.8430

Frequency spectrum (FFT)



Dominant frequency

2771.5 Hz

Prediction

Predicted condition: **Inner Race Fault** — confidence 100.0% (averaged over 1 window)

Estimated fault size: **0.021 inch (0.53 mm)** — confidence 100.0%

⚠ Motor needs inspection — Inner Race Fault detected

Prediction

Predicted condition: **Normal** — confidence 100.0% (averaged over 1 window)

✅ Motor healthy

Upload: *sample_normal.csv*

Run: `python3 -m streamlit run app.py`

- Demo files in data/samples/ were left out of training (unseen 3 HP data).
- All 6 demo samples: correct fault type and fault size.
- 48 kHz sample: set rate to 48000 → converted, Normal 100%.
- Unfamiliar input (e.g. 100× louder signal) → "Result uncertain".

Upload: *sample_inner_race_fault_0_021in.csv*

Advantages



Type and severity

Names the damaged part and estimates the crack size, not just "faulty".



Noise robust

Keeps 92.5% accuracy even when noise is as strong as the signal.



Knows when it's unsure

Flags unfamiliar or low-confidence signals instead of guessing.



Any sampling rate

Signals recorded at other rates are converted to 12 kHz automatically.



Lightweight

Random Forest on 14 features: trains in about 20 s on a laptop, no GPU.



Explainable

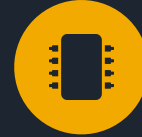
Each feature (RMS, kurtosis, band energy) has a physical meaning.

Limitations and future scope

CURRENT LIMITATIONS

- Software prototype on recorded lab data from one test rig; no hardware or real-time monitoring yet
- Does not generalise to unseen crack sizes or a different sensor position
- Only bearing faults; misalignment, unbalance and looseness are not classified
- User must enter the correct sampling rate

FUTURE SCOPE



Accelerometer + ESP32

Capture vibration from a real motor for real-time monitoring.



IoT / cloud dashboard

Remote monitoring, alerts and trend history for many motors.



Envelope analysis

Physics-based features at bearing defect frequencies to generalise to new sizes and sensors.



Real industrial data

Train on factory data and more fault types.

Thank you

Questions?

98.82%

fault type at an unseen load

99.25%

fault size at an unseen load

92.5%

accuracy with heavy noise